

Aplicación Java con MongoDB y arquitectura en capas



{ name: mongo, type: DB }

Descripción de proyecto

Para la creación de un proyecto de mongo he decidido usar como base mi proyecto TFG que estamos desarrollando en varias asignaturas.

Este consiste en mejorar el hábito lector, la velocidad y la comprensión lectora de sus usuarios. Se enfoca en estudiantes, utilizando la gamificación (logros, niveles) y la Inteligencia Artificial (IA) para adaptar el contenido y los ejercicios al nivel de competencia de cada usuario.

Nuestro sistema gestionará diferentes roles (Estudiantes, Docentes, Familias) y almacena la información clave: lecturas disponibles, actividades de evaluación, progreso individual de los estudiantes y las asignaciones de tareas que los docentes establecen.

El modelo que vamos a utilizar es el siguiente:

1. Documento Principal: Usuario

El documento **Usuario** centraliza toda la información del perfil de un individuo en la plataforma (estudiante, docente o familiar).

2. Lectura

El documento **Lectura** describe un texto o fragmento de contenido didáctico disponible en el catálogo de la aplicación.

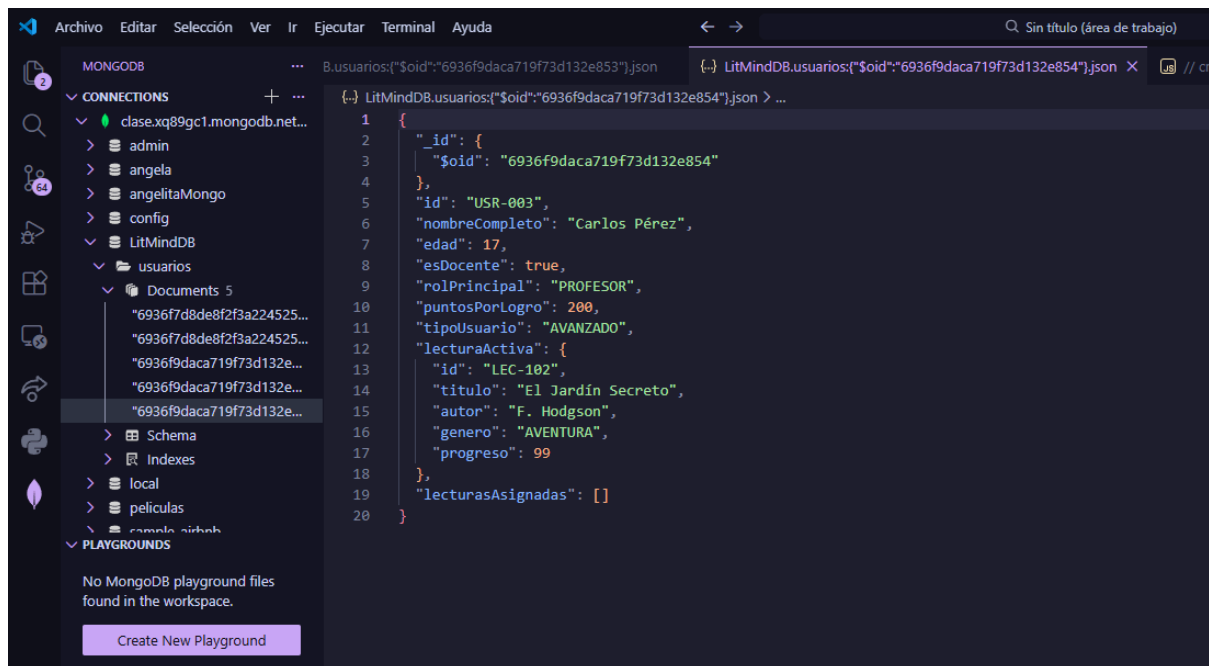
3. Asignación

El documento **Asignación** es el nexo entre un Docente (Usuario) y un conjunto de Lecturas, representando una tarea específica asignada a un grupo de estudiantes.

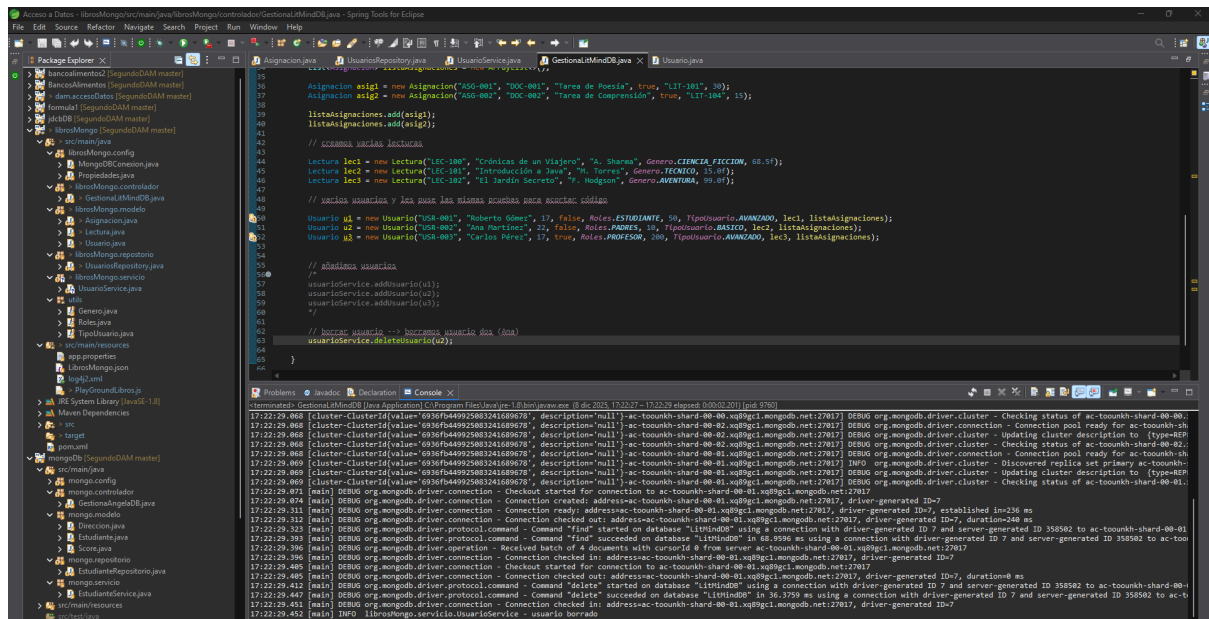
Funcionamiento:

- añadir usuario:

```
20
21
22 public class GestionaLitMindDB {
23
24     private static final Logger logger = LogManager.getLogger(UserService.class);
25
26     public static void main(String[] args) {
27         MongoClient conexion = new MongoClient();
28         MongoDatabase db = conexion.getDatabase();
29
30         // Instanciamos los objetos
31         UserService userService = new UserService(db);
32
33         // creamos una lista de asignaciones para realizamos varias para el mismo usuario
34         List<Asignacion> listaAsignaciones = new ArrayList<>();
35
36         Asignacion asign1 = new Asignacion("ASO-001", "DOC-001", "Tarea de Poesía", true, "LIT-101", 30);
37         Asignacion asign2 = new Asignacion("ASO-002", "DOC-002", "Tarea de Comprensión", true, "LIT-104", 15);
38         listaAsignaciones.add(asign1);
39         listaAsignaciones.add(asign2);
40
41         // creamos varias lecturas
42
43         Lectura lec1 = new Lectura("LEC-001", "Crónicas de un Viajero", "H. Sharn", Genero.CIENCIA_FICCION, 68.5f);
44         Lectura lec2 = new Lectura("LEC-101", "Introducción a Java", "M. Torres", Genero.TECNICO, 15.8f);
45         Lectura lec3 = new Lectura("LEC-102", "El Jardín Secreto", "F. Hodgson", Genero.AVENTURA, 99.8f);
46
47         // vamos a crear y los ponemos las mismas pruebas para algunos códigos
48
49         Usuario u1 = new Usuario("USR-001", "Roberto Gómez", 17, false, Roles.ESTUDIANTE, 10, TipoUsuario.AVANZADO, lec1, listaAsignaciones);
50         Usuario u2 = new Usuario("USR-002", "Ana Martínez", 22, false, Roles.PROFESOR, 10, TipoUsuario.BASICO, lec2, listaAsignaciones);
51         Usuario u3 = new Usuario("USR-003", "Carlos Pérez", 17, true, Roles.PROFESOR, 10, TipoUsuario.AVANZADO, lec3, listaAsignaciones);
52
53         // añadimos usuarios
54         userService.addUsuario(u1);
55         userService.addUsuario(u2);
56         userService.addUsuario(u3);
57     }
58 }
59
60
61
62
63
64
```



- borrar usuario:



```
Asignacion.java  UsuariosRepository.java  UsuarioService.java  GestionarLitMindDB.java  Usuario.java

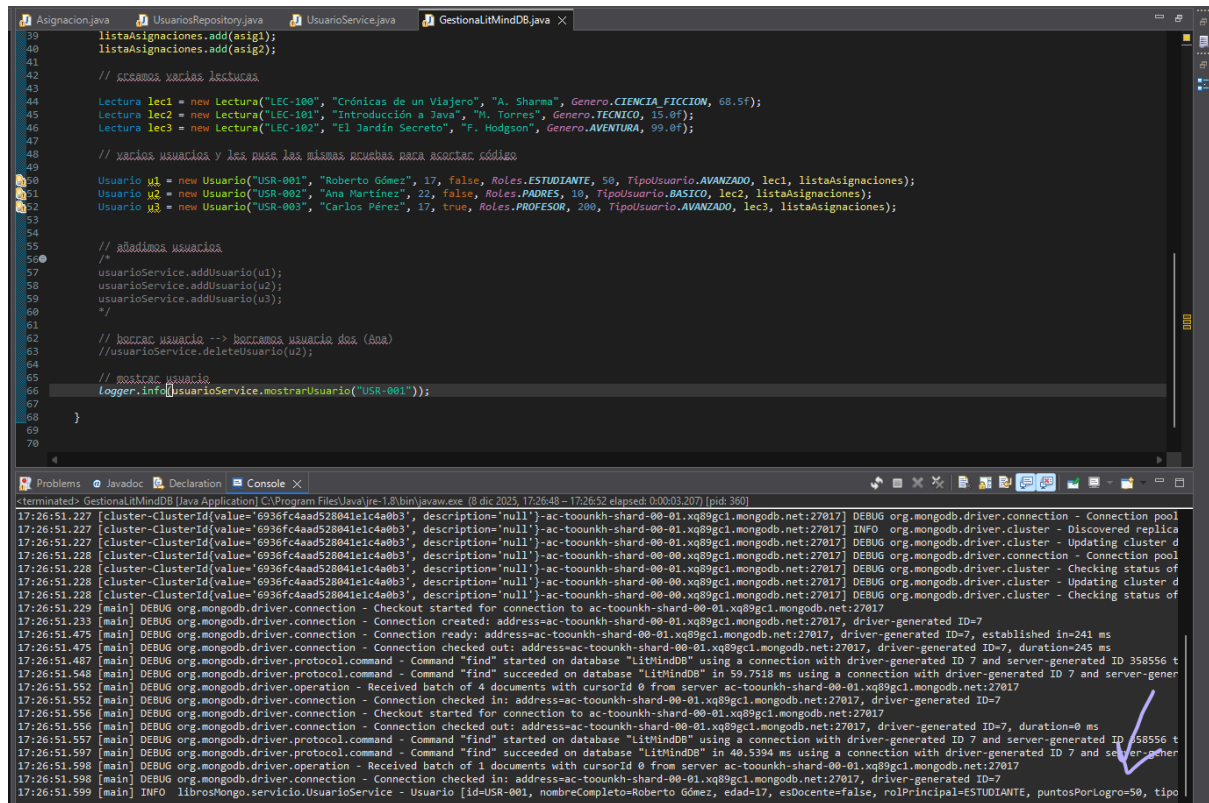
// ...

// borrar usuario --> borrar usuario dos (dos)
usuarioService.deleteUsuario(u2);

// ...

Problems  |  Javadoc  |  Declaration  |  Console  |
-----
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Checking status of ac-toonkh-shard-00-01.
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-02.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Connection pool ready for ac-toonkh-sh
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-02.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Updating cluster description to {type=REP
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-02.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Checking status of ac-toonkh-shard-00-02.
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.connection - Connection pool ready for ac-toonkh-sh
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] INFO org.mongodb.driver.cluster - Discovered replica set primary ac-toonkh
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Updating cluster description to {type=REP
17:22:29.468 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Checking status of ac-toonkh-shard-00-01.
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection created: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=236 ms
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=240 ms
17:22:29.471 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "litminddb" using a connection with driver-generated ID 7 and server-generated ID 358582 to ac-toonkh-shard-00-01
17:22:29.471 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "litminddb" in 68.5586 ms using a connection with driver-generated ID 7 and server-generated ID 358582 to ac-to
17:22:29.471 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=245 ms
17:22:29.471 [main] DEBUG org.mongodb.driver.protocol.command - Command "delete" started on database "litminddb" using a connection with driver-generated ID 7 and server-generated ID 358582 to ac-toonkh-shard-00-01
17:22:29.471 [main] DEBUG org.mongodb.driver.protocol.command - Command "delete" succeeded on database "litminddb" in 36.3759 ms using a connection with driver-generated ID 7 and server-generated ID 358582 to ac-b
17:22:29.471 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
17:22:29.471 [main] INFO librosMongo.servicio.UsuarioService - usuario borrado
```

- mostrar usuario:



```
Asignacion.java  UsuariosRepository.java  UsuarioService.java  GestionarLitMindDB.java

// ...

// mostrar usuario
logger.info(usuarioService.mostrarUsuario("USR-001"));

// ...

Problems  |  Javadoc  |  Declaration  |  Console  |
-----
17:26:51.227 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.connection - Connection pool
17:26:51.227 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] INFO org.mongodb.driver.cluster - Discovered replica
17:26:51.227 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Updating cluster d
17:26:51.228 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.connection - Connection pool
17:26:51.228 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Checking status of
17:26:51.228 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-00.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Updating cluster d
17:26:51.228 [cluster-ClusterId{value='6936f44ad528041e1c4a0b3', description='null'}-ac-toonkh-shard-00-00.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Checking status of
17:26:51.229 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:26:51.233 [main] DEBUG org.mongodb.driver.connection - Connection created: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
17:26:51.475 [main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=241 ms
17:26:51.475 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=245 ms
17:26:51.487 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "litminddb" using a connection with driver-generated ID 7 and server-generated ID 358556 t
17:26:51.597 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "litminddb" in 40.5394 ms using a connection with driver-generated ID 7 and server-gener
17:26:51.552 [main] DEBUG org.mongodb.driver.operation - Received batch of 1 documents with cursorId 0 from server ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:26:51.552 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
17:26:51.556 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:26:51.556 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms
17:26:51.557 [main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=245 ms
17:26:51.597 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "litminddb" using a connection with driver-generated ID 7 and server-generated ID 358556 t
17:26:51.598 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "litminddb" in 59.7518 ms using a connection with driver-generated ID 7 and server-gener
17:26:51.598 [main] DEBUG org.mongodb.driver.connection - Received batch of 1 documents with cursorId 0 from server ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017
17:26:51.598 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toonkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
17:26:51.599 [main] INFO librosMongo.servicio.UsuarioService - Usuario [id=USR-001, nombreCompleto=Roberto Gómez, edad=17, esDocente=false, rolPrincipal=ESTUDIANTE, puntosPorLogro=50, tipo
```

- actualizar usuario (cambiamos el valor que deseamos)

```
Asignacion.java  UsuariosRepository.java  UsuarioService.java  GestionaLitMindDB.java  Usuario.java
43
44     Lectura lec1 = new Lectura("LEC-100", "Crónicas de un Viajero", "A. Sharma", Genero.CIENCIA_FICCION, 68.5f);
45     Lectura lec2 = new Lectura("LEC-101", "Introducción a Java", "M. Torres", Genero.TECNICO, 15.0f);
46     Lectura lec3 = new Lectura("LEC-102", "El Jardín Secreto", "F. Hodgson", Genero.AVENTURA, 99.0f);
47
48     // varios usuarios y les pasa las mismas pruebas para acortar código
49
50     Usuario u1 = new Usuario("USR-001", "Roberto Gómez", 17, false, Roles.ESTUDIANTE, 50, TipoUsuario.AVANZADO, lec1, listaAsignaciones);
51     Usuario u2 = new Usuario("USR-002", "Ana Martínez", 22, false, Roles.PADRES, 10, TipoUsuario.BASICO, lec2, listaAsignaciones);
52     Usuario u3 = new Usuario("USR-003", "Carlos Pérez", 17, true, Roles.PROFESOR, 200, TipoUsuario.AVANZADO, lec3, listaAsignaciones);
53
54
55     // añadimos usuarios
56     /*
57     usuarioService.addUsuario(u1);
58     usuarioService.addUsuario(u2);
59     usuarioService.addUsuario(u3);
60     */
61
62     // borrar usuario --> borramos usuario de Ana
63     //usuarioService.deleteUsuario(u2);
64
65     // mostrar usuario
66     // logger.info(usuarioService.mostrarUsuario("USR-001"));
67
68     // actualizar usuario --> hacemos que cambie el que mostramos antes
69     // creamos los valores nuevos y luego lo cambiamos por el usuario con el id --> USR-1001
70
71     u3.setNombreCompleto("Pedro Pérez");
72     usuarioService.updateUsuario(u3);
73
74 }
75
76 }
77
78 }
```

```
<terminated> GestionaLitMindDB [Java Application] C:\Program Files\Java\jre-1.8\bin\javaw.exe (8 dic 2025, 17:59:59 - 18:00:02 elapsed: 0:00:02.182) [pid: 9844]
18:00:01.332 [Cluster-ClusterId{value='69370410db63309d03dbaf2', description='null'}-ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017] INFO org.mongodb.driver.cluster - Discovered replica
18:00:01.332 [Cluster-ClusterId{value='69370410db63309d03dbaf2', description='null'}-ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Updating cluster d
18:00:01.332 [Cluster-ClusterId{value='69370410db63309d03dbaf2', description='null'}-ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017] DEBUG org.mongodb.driver.cluster - Checking status of
18:00:01.333 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:00:01.336 [main] DEBUG org.mongodb.driver.connection - Connection created: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:00:01.529 [main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=192 ms
18:00:01.531 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=196 ms
18:00:01.544 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 358960 t
18:00:01.590 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 42.4259 ms using a connection with driver-generated ID 7 and server-gener
18:00:01.595 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:00:01.596 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:00:01.607 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:00:01.608 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms
18:00:01.611 [main] DEBUG org.mongodb.driver.protocol.command - Command "findAndModify" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID
18:00:01.652 [main] DEBUG org.mongodb.driver.protocol.command - Command "findAndModify" succeeded on database "LitMindDB" in 41.0866 ms using a connection with driver-generated ID 7 and ser
18:00:01.653 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:00:01.653 [main] INFO librosMongo.servicio.UsuarioService - usuario se actualizó
```

Archivo Editar Selección Ver Ir Ejecutar Terminal Ayuda

MONGODB

CONNECTIONS

- clase.xq89gcl.mongodb.net...
- admin
- angela
- angelitaMongo
- config
- LitMindDB
 - usuarios
 - Documents 4
 - "6936f7d8de8f2f3a224525...
 - "6936f7d8de8f2f3a224525...
 - "6936f9daca719f73d132e...
 - "6936f9daca719f73d132e...
- Schema
- Indexes
- local
- peliculas
- sample_airbnb
- sample_analyzer

PLAYGROUNDS

No MongoDB playground files found in the workspace.

Create New Playground

```
{-} LitMindDB.usuarios:{"$oid":"6936f9daca719f73d132e854"}json X
{-} LitMindDB.usuarios:{"$oid":"6936f9daca719f73d132e854"}json > ...
1 {
2   "_id": {
3     "$oid": "6936f9daca719f73d132e854"
4   },
5   "id": "USR-003",
6   "nombreCompleto": "Pedro Pérez",
7   "edad": 17,
8   "esDocente": true,
9   "rolPrincipal": "PROFESOR",
10  "puntosPorLogro": 200,
11  "tipoUsuario": "AVANZADO",
12  "lecturaActiva": {
13    "id": "LEC-102",
14    "titulo": "El Jardín Secreto",
15    "autor": "F. Hodgson",
16    "genero": "AVENTURA",
17    "progreso": 99
18  },
19  "lecturasAsignadas": []
20 }
```

- filtramos por edad:

```
Asignacion.java | UsuariosRepository.java | UsuarioService.java | GestionaLitMindDB.java | Usuario.java
38
39 listaAsignaciones.add(asig1);
40 listaAsignaciones.add(asig2);
41
42 // creamos varias lecturas
43
44 Lectura lec1 = new Lectura("LEC-100", "Crónicas de un Viajero", "A. Sharma", Genero.CIENCIA_FICCION, 68.5f);
45 Lectura lec2 = new Lectura("LEC-101", "Introducción a Java", "M. Torres", Genero.TECNICO, 15.0f);
46 Lectura lec3 = new Lectura("LEC-102", "El Jardín Secreto", "F. Hodgson", Genero.AVENTURA, 99.0f);
47
48 // vamos a crear y a la vez a las mismas acciones para asociar códigos
49
50 Usuario u1 = new Usuario("USR-001", "Roberto Gómez", 17, false, Roles.ESTUDIANTE, 50, TipoUsuario.AVANZADO, lec1, listaAsignaciones);
51 Usuario u2 = new Usuario("USR-002", "Ana Martínez", 22, false, Roles.PADRES, 10, TipoUsuario.BASICO, lec2, listaAsignaciones);
52 Usuario u3 = new Usuario("USR-003", "Carlos Pérez", 17, true, Roles.PROFESOR, 200, TipoUsuario.AVANZADO, lec3, listaAsignaciones);
53
54
55 // añadimos usuarios
56
57 usuarioService.addUsuario(u1);
58 usuarioService.addUsuario(u2);
59 usuarioService.addUsuario(u3);
60
61
62 // borramos usuario --> borramos usuario Ana
63 // usuarioService.deleteUsuario(u2);
64
65 // mostramos usuario
66 // logger.info(usuarioService.mostrarUsuario("USR-001"));
67
68 // actualiza usuario --> elegimos que cambiar el que mostramos antes
69 // creamos los valores nuevos y luego lo cambiamos por el usuario con el id --> USR-1001
70
71 //u1.setNombreCompleto("Pedro Pérez");
72 usuarioService.updateUsuario(u3);
73
74 //logger.info(usuarioService.filtrarPorEdad(17));
75
76 }
77
78 }
79
80 }
```

Problems | Javadoc | Declaration | Console |

terminated: GestionaLitMindDB [Java Application] [C:\Program Files\Java\jre-1.8\bin\java.exe (8 dic 2025, 18:04:19 - 18:04:21 elapsed: 0:00:02.091) [pid: 27556]

18:04:20.775 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 358997 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.812 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 35.8009 ms using a connection with driver-generated ID 7 and server-generated ID 358997 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.815 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.816 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7

18:04:20.819 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.820 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms

18:04:20.821 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 358997 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.821 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 37.8794 ms using a connection with driver-generated ID 7 and server-generated ID 358997 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.859 [main] DEBUG org.mongodb.driver.operation - Received batch of 3 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:04:20.859 [main] DEBUG org.mongodb.driver.operation - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7

18:04:20.860 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7

18:04:20.860 [main] INFO librosMongo.servicio.UsuarioService - [Usuario [id=USR-1002, nombreCompleto=Roberto Gómez Piñol, edad=17, esDocente=false, rolPrincipal=ESTUDIANTE, puntosPorLogro=10, tipoUsuario=BASICO

- ordenamos por nombre:

```
Asignacion.java | UsuariosRepository.java | UsuarioService.java | GestionaLitMindDB.java | Usuario.java
49
50 Usuario u1 = new Usuario("USR-001", "Roberto Gómez", 17, false, Roles.ESTUDIANTE, 50, TipoUsuario.AVANZADO, lec1, listaAsignaciones);
51 Usuario u2 = new Usuario("USR-002", "Ana Martínez", 22, false, Roles.PADRES, 10, TipoUsuario.BASICO, lec2, listaAsignaciones);
52 Usuario u3 = new Usuario("USR-003", "Carlos Pérez", 17, true, Roles.PROFESOR, 200, TipoUsuario.AVANZADO, lec3, listaAsignaciones);
53
54
55 // añadimos usuarios
56
57 usuarioService.addUsuario(u1);
58 usuarioService.addUsuario(u2);
59 usuarioService.addUsuario(u3);
60
61
62 // borramos usuario --> borramos usuario Ana
63 // usuarioService.deleteUsuario(u2);
64
65 // mostramos usuario
66 // logger.info(usuarioService.mostrarUsuario("USR-001"));
67
68 // actualiza usuario --> elegimos que cambiar el que mostramos antes
69 // creamos los valores nuevos y luego lo cambiamos por el usuario con el id --> USR-1001
70
71 //u1.setNombreCompleto("Pedro Pérez");
72 usuarioService.updateUsuario(u3);
73
74 // filtramos por edad
75 // logger.info(usuarioService.filtrarPorEdad(17));
76
77 // ordenamos por nombre
78 // logger.info(usuarioService.ordenarPorNombre());
79
80 }
81
82 }
83
84 }
85
86 }
```

Problems | Javadoc | Declaration | Console |

terminated: GestionaLitMindDB [Java Application] [C:\Program Files\Java\jre-1.8\bin\java.exe (8 dic 2025, 18:06:32 - 18:06:34 elapsed: 0:00:02.153) [pid: 24016]

18:06:34.131 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.132 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7

18:06:34.136 [main] DEBUG org.mongodb.driver.connection - Checkout started for connection to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.136 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms

18:06:34.137 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 359036 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.137 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 37.466 ms using a connection with driver-generated ID 7 and server-generated ID 359036 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.175 [main] DEBUG org.mongodb.driver.operation - Received batch of 3 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.175 [main] DEBUG org.mongodb.driver.operation - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7

18:06:34.176 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms

18:06:34.176 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=0 ms

18:06:34.177 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 359036 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.214 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 36.5188 ms using a connection with driver-generated ID 7 and server-generated ID 359036 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.217 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017

18:06:34.217 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7

18:06:34.218 [main] INFO librosMongo.servicio.UsuarioService - [Usuario [id=USR-1001, nombreCompleto=Ana Ríos Martínez, edad=15, esDocente=false, rolPrincipal=ESTUDIANTE, puntosPorLogro=80, tipoUsuario=AVANZADO

- filtramos por tipo de usuario

```
Asignacion.java UsuariosRepository.java UsuarioService.java GestionaLitMindDB.java Usuario.java
49
50 Usuario u1 = new Usuario("USR-001", "Roberto Gómez", 17, false, Roles.ESTUDIANTE, 50, TipoUsuario.AVANZADO, lect, listaAsignaciones);
51 Usuario u2 = new Usuario("USR-002", "Ana Martínez", 22, false, Roles.PADRES, 10, TipoUsuario.BASICO, lec2, listaAsignaciones);
52 Usuario u3 = new Usuario("USR-003", "Carlos Pérez", 17, true, Roles.PROFESOR, 200, TipoUsuario.AVANZADO, lec3, listaAsignaciones);
53
54
55 // añadimos usuarios
56 /*
57 usuarioService.addUsuario(u1);
58 usuarioService.addUsuario(u2);
59 usuarioService.addUsuario(u3);
60 */
61
62 // borramos usuario --> borramos usuario dos (Ana)
63 //usuarioService.deleteUsuario(u2);
64
65 // mostramos usuario
66 // logger.info(usuarioService.mostrarUsuario("USR-001"));
67
68 // actualiza usuario --> elegimos que cambiar el que mostramos antes
69 // creamos los valores nuevos y luego los sustituimos por el usuario con el id --> USR-1001
70
71 /*u3.setNombreCompleto("Pedro Pérez");
72 usuarioService.updateUsuario(u3);*/
73
74 // filtramos por edad
75 // logger.info(usuarioService.filtrarPorEdad(17));
76
77 // ordenamos por nombre
78 // logger.info(usuarioService.ordenarPorNombre());
79
80 // filtramos tipo usuario
81 // logger.info(usuarioService.filtrarTipoUsuario(TipoUsuario.AVANZADO));
82
83 }
84
85 }
86 }
```

↓

```
Problems Javadoc Declaration Console
terminated: GestionaLitMindDB [Java Application] C:\Program Files\Java\jre-1.8\bin\java.exe (8 dic 2025, 18:19:32 - 18:19:34 elapsed: 0:00:02.130) [pid: 28736]
[Cluster-ClusterId(value="693708a50a1e2d767955d5f5", description="null")] ac-toounkh-shard-00-00.xq89gcl.mongodb.net:27017 DEBUG org.mongodb.driver.cluster - Checking status of ac-toounkh-shard-00-00.xq89gcl.mongodb.net:27017
[main] DEBUG org.mongodb.driver.connection - Connection created: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
[main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=222 ms
[main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=226 ms
[main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 359177 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
[main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
[main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
[main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
[main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 359177 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
[main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 33.3487 ms using a connection with driver-generated ID 7 and server-generated ID 359177 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
[main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
[main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
[main] INFO librosMongo.servicio.UsuarioService - [Usuario [id=USR-1001, nombreCompleto=Elena Rios Martinez, edad=15, esDocente=false, rolPrincipal=ESTUDIANTE, puntosPorLogro=80, tipoUsuario=AVANZADO, lecturaActiva=...
```

- ordenamos por puntos de forma descendente:

```
Asignacion.java UsuariosRepository.java UsuarioService.java GestionaLitMindDB.java Usuario.java
53
54 Usuario u3 = new Usuario("USR-003", "Carlos Pérez", 17, true, Roles.PROFESOR, 200, TipoUsuario.AVANZADO, lec3, listaAsignaciones);
55
56 // añadimos usuarios
57 /*
58 usuarioService.addUsuario(u1);
59 usuarioService.addUsuario(u2);
60 usuarioService.addUsuario(u3);
61 */
62
63 // borramos usuario --> borramos usuario dos (Ana)
64 //usuarioService.deleteUsuario(u2);
65
66 // mostramos usuario
67 // logger.info(usuarioService.mostrarUsuario("USR-001"));
68
69 // actualiza usuario --> elegimos que cambiar el que mostramos antes
70 // creamos los valores nuevos y luego los sustituimos por el usuario con el id --> USR-1001
71
72 /*u3.setNombreCompleto("Pedro Pérez");
73 usuarioService.updateUsuario(u3);*/
74
75 // filtramos por edad
76 // logger.info(usuarioService.filtrarPorEdad(17));
77
78 // ordenamos por nombre
79 // logger.info(usuarioService.ordenarPorNombre());
80
81 // filtramos tipo usuario
82 // logger.info(usuarioService.filtrarTipoUsuario(TipoUsuario.AVANZADO));
83
84 // ordenamos por puntos descendente
85 // logger.info(usuarioService.ordenaPuntosLogros());
86
87 }
88
89 }
```

```
Problems Javadoc Declaration Console
terminated: GestionaLitMindDB [Java Application] C:\Program Files\Java\jre-1.8\bin\java.exe (8 dic 2025, 18:24:52 - 18:24:54 elapsed: 0:00:02.162) [pid: 21984]
18:24:54.008 [Cluster-ClusterId(value="693708a50a1e2d767955d5f5", description="null")] ac-toounkh-shard-00-02.xq89gcl.mongodb.net:27017 DEBUG org.mongodb.driver.cluster - Checking status of ac-toounkh-shard-00-02.xq89gcl.mongodb.net:27017
18:24:54.009 [main] DEBUG org.mongodb.driver.connection - Connection created: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:24:54.012 [main] DEBUG org.mongodb.driver.connection - Connection ready: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, established in=213 ms
18:24:54.026 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7, duration=217 ms
18:24:54.237 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 359245 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:24:54.274 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:24:54.277 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:24:54.278 [main] DEBUG org.mongodb.driver.connection - Connection checked out: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:24:54.281 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" started on database "LitMindDB" using a connection with driver-generated ID 7 and server-generated ID 359245 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:24:54.281 [main] DEBUG org.mongodb.driver.protocol.command - Command "find" succeeded on database "LitMindDB" in 82.4496 ms using a connection with driver-generated ID 7 and server-generated ID 359245 to ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:24:54.274 [main] DEBUG org.mongodb.driver.operation - Received batch of 4 documents with cursorId 0 from server ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017
18:24:54.366 [main] DEBUG org.mongodb.driver.connection - Connection checked in: address=ac-toounkh-shard-00-01.xq89gcl.mongodb.net:27017, driver-generated ID=7
18:24:54.366 [main] INFO librosMongo.servicio.UsuarioService - [Usuario [id=USR-003, nombreCompleto=Pedro Pérez, edad=17, esDocente=true, rolPrincipal=PROFESOR, puntosPorLogro=200, tipoUsuario=AVANZADO, lecturaActiva=...
```